

Bitwise differential trail through **QALQAN** found by SAT

CiVerLy

August 4, 2026

1 QALQAN

Maximal differential probability: 2^{-93}

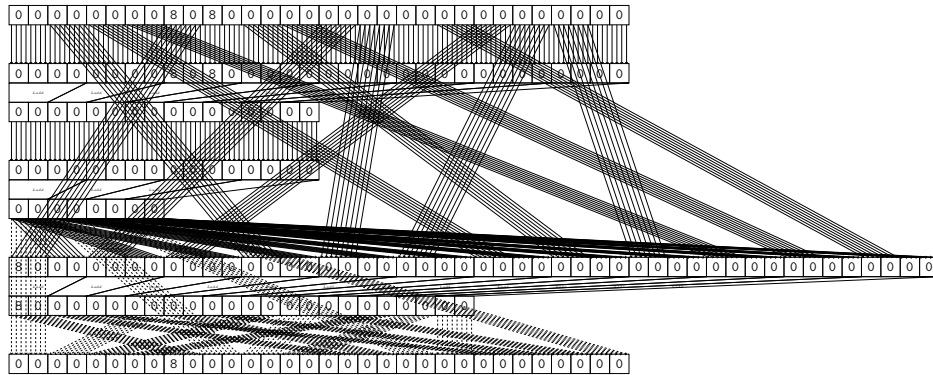
0	0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>RcapAddressOut</i>																															
0	0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	f	0	f	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>SDataLeger</i>																															
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>L</i>																															
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>QalqanRound</i>																															
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>QalqanRound</i>																															
0	0	0	0	0	0	0	0	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>QalqanRound</i>																															
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>QalqanRound</i>																															
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>QalqanRound</i>																															
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0	0	0	0	0	0

2 SBoxLayer

Maximal differential probability: $2^{-12.0}$ [illegible]

3 L

Maximal differential probability: $2^{-0.0}$



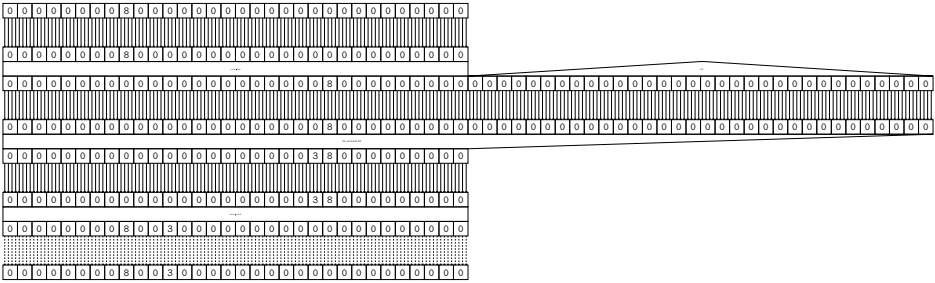
4 QalqanRound

Maximal differential probability: $2^{-19.0}$

0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AES128																												
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2RoundLayer																												
0	0	0	0	0	0	0	0	4	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	4	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
L																												
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

5 Add128

Maximal differential probability: $2^{-3.0}$



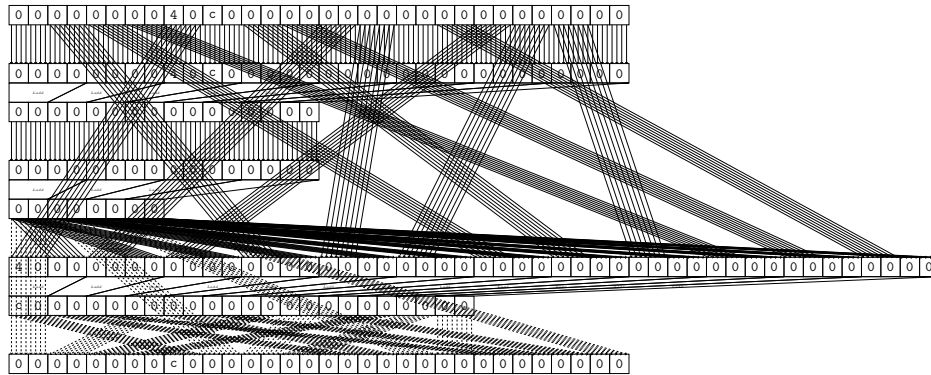
6 SBoxLayer

Maximal differential probability: $2^{-14.0}$

0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	4	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	4	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

7 L

Maximal differential probability: $2^{-2.0}$



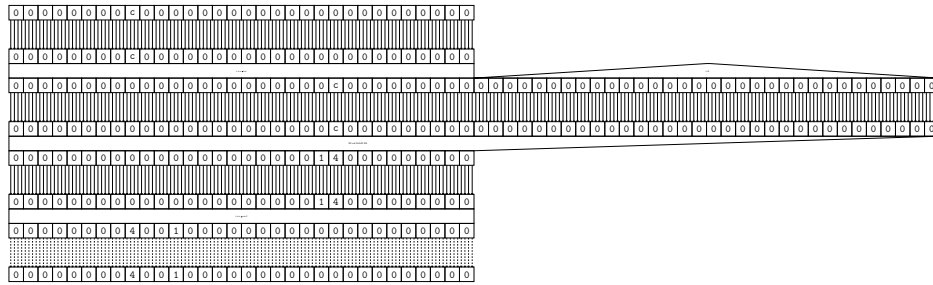
8 QalqanRound

Maximal differential probability: $2^{-19.0}$

0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	c	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AES128																												
0	0	0	0	0	0	0	0	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2RoundLayer																												
0	0	0	0	0	0	0	0	8	1	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	1	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
L																												
0	0	0	0	0	0	0	0	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

9 Add128

Maximal differential probability: $2^{-3.0}$



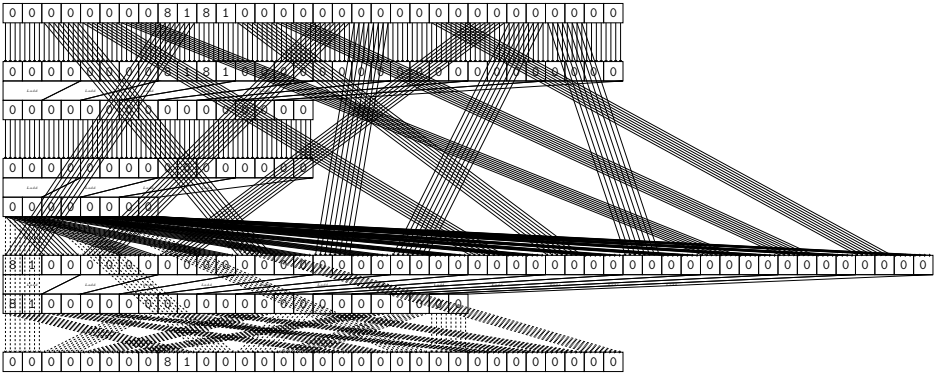
10 SBoxLayer

Maximal differential probability: 2^{-14.0}

0	0	0	0	0	0	0	0	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	8	1	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	1	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

11 L

Maximal differential probability: $2^{-2.0}$



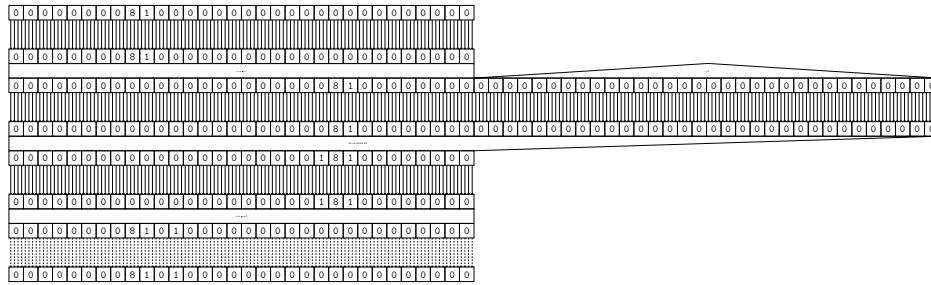
12 QalqanRound

Maximal differential probability: $2^{-17.0}$

0	0	0	0	0	0	0	0	0	0	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	8	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
AES128																															
0	0	0	0	0	0	0	0	0	8	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	8	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
2RoundLayer																															
0	0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
L																															
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

13 Add128

Maximal differential probability: $2^{-3.0}$



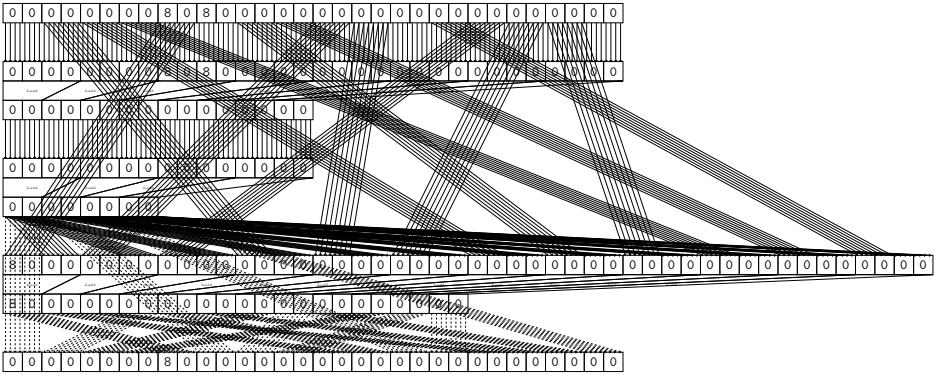
14 SBoxLayer

Maximal differential probability: $2^{-14.0}$

0	0	0	0	0	0	0	0	8	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	<i>SBox</i>	
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

15 L

Maximal differential probability: $2^{-0.0}$

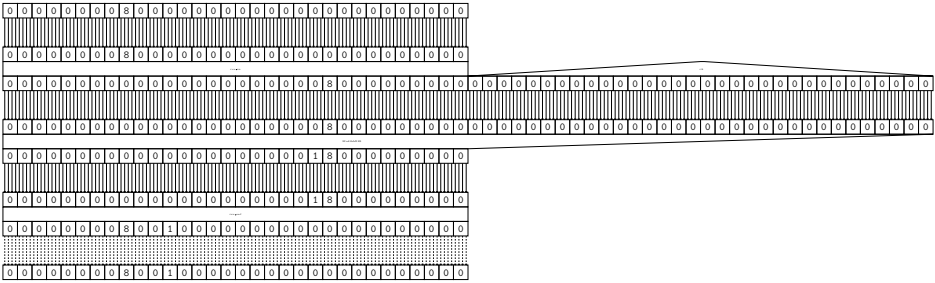


16 QalqanRound

Maximal differential probability: $2^{-18.0}$ [illegible]

17 Add128

Maximal differential probability: $2^{-2.0}$

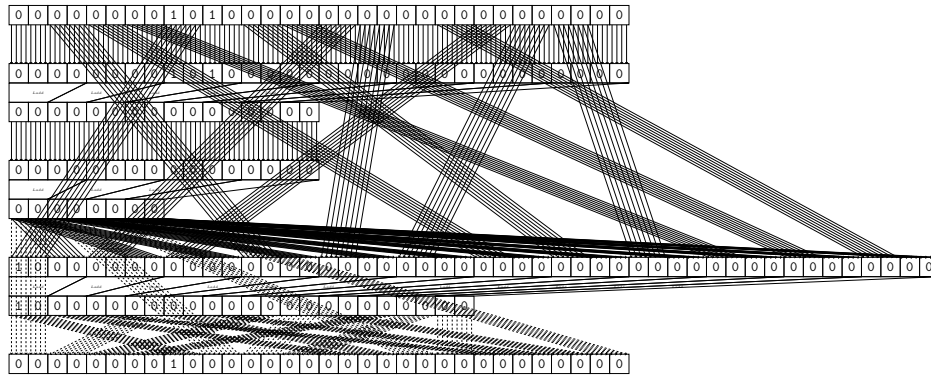


18 SBoxLayer

Maximal differential probability: $2^{-14.0}$ [illegible]

19 L

Maximal differential probability: $2^{-2.0}$



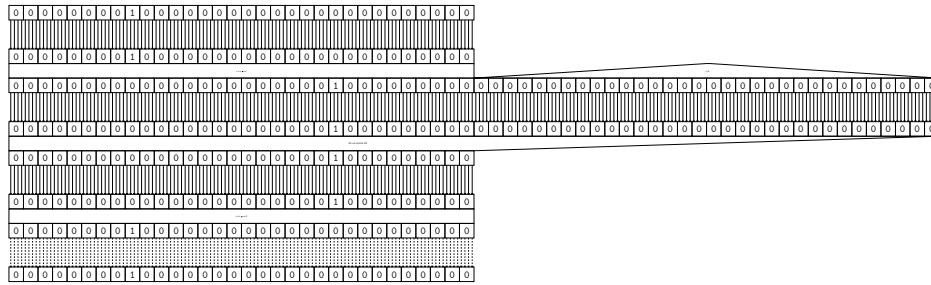
20 QalqanRound

Maximal differential probability: $2^{-8.0}$

0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AES128																											
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2RoundKey																											
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
L																											
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0
0	0	8	0	0	0	0	0	8	0	8	0	0	0	0	0	0	0	8	0	0	0	0	0	0	8	0	0

21 Add128

Maximal differential probability: $2^{-1.0}$



22 SBoxLayer

Maximal differential probability: $2^{-7.0}$

0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox	SBox
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

23 L

Maximal differential probability: $2^{-0.0}$

